Series circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use current continuity and voltage drops in a single-path circuit.

SUCCESS CRITERIA - I can...

1. Identify a single current path.
2. Add series resistances.
3. Check that voltage drops add to the supply voltage.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “series” and “current” to explain your answer. Sentence starters are optional.

series: connected along one unbranched current path

current: rate of charge flow

resistance: voltage divided by current for a stated operating condition

01 Warm-up: bring back an earlier idea

Two series components have drops **3 V** and **5 V**. A charge of **7 C** passes through each. **Find each energy transfer** and the total.

Word help: charge = a property responsible for electric interactions

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

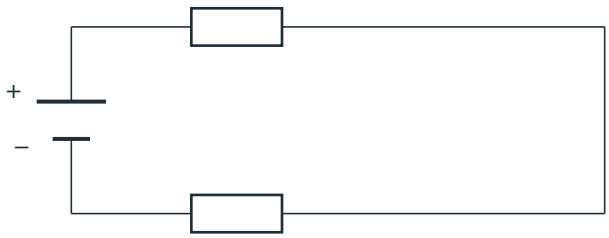
02 Worked example: explain the prediction

Reference: $R_{total} = R_1 + R_2$; $I_1 = I_2$; $V_{total} = V_1 + V_2$. Series resistors share one unbranched current path.

Worked reasoning: Two series resistors **2** and **4 ohms** on a **12 V** ideal supply have total **6 ohms** and current **2 A**. Drops are **4 V** and **8 V**; they add to **12 V**.

Your turn: Draw the loop and **label current** and both voltage drops.

Sentence starter: In my work, I used … to show … . I labeled it … .



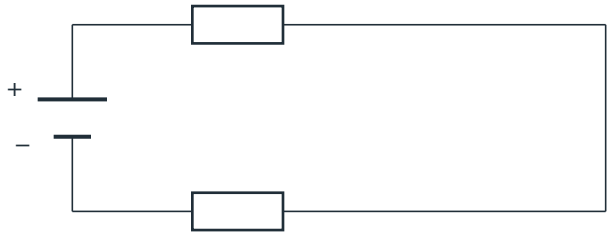
Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Trace the only current path in the series diagram. If either resistor connection opens, **what happens** to the current?

Sentence starter: In my work, I used … to show … . I labeled it … .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

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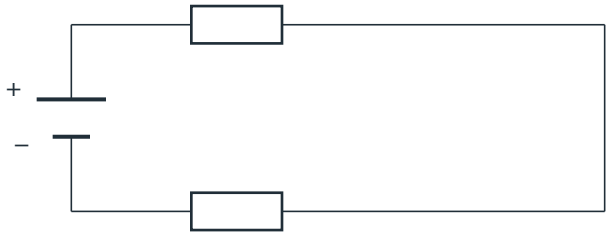
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **12 V** lab supply powers **10 Ω** and **20 Ω** resistors in series.
1. **Label the circuit**, then **find total resistance**, current and both voltage drops.
2. Use the drops to **check the loop's energy-per-charge balance**.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

05 Explain your reasoning

A student assigns more current to the smaller series resistor. **Explain why this would imply** charge accumulating between resistors.

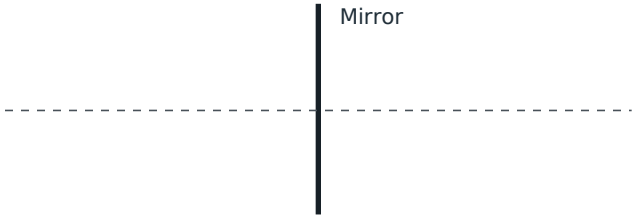
Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An object moves **0.4 m** toward a fixed plane mirror. **Find image movement** and the change in object-image separation.
Mark the object and image positions before and after the change.

Sentence starter: My design includes … . It does what is needed by … .



Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

07 Exit check

1. **Add a third equal resistor** to two equal series resistors at the same supply voltage.
2. **Compare new current** with original current.
3. **Label the circuit** and **show the current path or** paths.

Sentence starter: Both … . They differ in … because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Three series resistors **2**, **3**, R ohms draw **2 A** from **20 V**.
1. **Find R** and power in each resistor.
2. **Label the circuit** and **show the current path or** paths.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

Marzano check - circle: ① ② ③